

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

September 2025 – May 2027 (Expected)

Amherst, MA

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend & APIs	Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, authentication, rate limiting, audit logging
Cloud & DevOps	AWS/GCP basics, Docker, Git, GitHub Actions, CI/CD, Linux, Cloudflare Workers
Data & Systems	Spark, PySpark, Databricks, Spark UI profiling, multi-tenant architecture
Databases & AI	MongoDB, PostgreSQL, Redis, SQLite, OpenAI / Groq APIs, RAG, LLM agents, PyTorch, Claude Code / Cursor / Copilot

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned the **multi-tenant NHS eTriage backend** end-to-end for **8+ UK clients** (Node.js, MongoDB), building software components integrated into Mayden's iaptus platform via REST APIs and aggregation pipelines to process **10,000+ monthly submissions** and contribute to **\$340K+** in revenue.
- Architected a full-stack **internal annotation platform (React, Node.js, MongoDB)** used daily by the AI team to label data for production models — internal tooling that replaced spreadsheet workflows with secure role-based REST APIs, cutting manual operations by **100+ hours/month** and improving labeling throughput by **60%**.
- Drove backend reliability and code quality across the eTriage stack: shipped maintainable auth and rate-limiting hardening, added integration tests for regression protection, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and shipped **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

github.com/Deva-1903/ciir_agentic_search

- Designed a multi-stage Python/FastAPI service that decomposes open-ended entity discovery into typed retrieval planning, page rerank, evidence-regime classification, deterministic extraction with LLM fallback, and output verification — a staged pipeline rather than a single opaque LLM call — returning cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

- Drove an ambiguous, previously-unsolved end-to-end build of a **cross-platform desktop application** integrating camera, microphone, frontend UI, and backend services into an on-device pipeline (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS) — designed, integrated, and shipped under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling video and audio streams locally on device.

Spark ETL Performance Optimization

Apache Spark, PySpark, Databricks, SQL, Python

github.com/Deva-1903/Spark-ETL-Optimization

- Optimized a Spark ETL pipeline on the NYC TLC taxi dataset using broadcast-join tuning, caching, partitioning, adaptive Spark settings, column pruning, and precomputed fields to reduce shuffle volume and wall-clock runtime.
- Used **Spark UI** stage-level metrics to evaluate complex data pipelines and resolve shuffle, skew, and runtime bottlenecks, then documented a repeatable optimization workflow comparing naive vs. tuned runs.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023** — applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.